# Phase 1 Completion Report: Project Foundation and Infrastructure Skeleton

## 1. Strategic Mission and Objective

The core mission of the AI Knowledge Base Evaluation Platform is to architect a production-grade system that meets the rigorous engineering standards of the 2026 AI job market. This platform transcends the saturated "simple chatbot" market by moving beyond subjective "vibe checks" toward a rigorous, data-driven engineering cycle. By establishing a robust infrastructure skeleton, we shift from a conceptual blueprint to a professional system capable of systematic measurement and continuous optimisation of Retrieval-Augmented Generation (RAG) performance.**Production-Grade (Senior-Level) vs. Standard Portfolio (Entry-Level)**

- **Standard Portfolio Anti-Pattern:** A linear path of Document Upload → Chat Interface → AI Answer. This relies on subjective assessment and lacks transparency into the retrieval quality.
- **Production-Grade Approach:** A closed-loop cycle of Document Upload → Structured Retrieval → AI Answer → Automated Evaluation → Performance Metrics → Iterative Experimentation. This approach treats AI output as a measurable engineering variable.Phase 1 focus has been the transition to a technical foundation that supports the **Ragas** and **Langfuse** integrations required for Phase 7 and 9. By synchronising environment constraints and deployment pipelines from the first commit, we eliminate the "works on my machine" fallacy and ensure the system is ready for high-concurrency LLM streaming.

## 2. Technical Infrastructure and Stack Validation

The technology stack has been meticulously selected to signal professional-grade technical judgment, prioritising scalability, type safety, and edge-runtime compatibility.

### Core Technology Stack

Component,Selection (Tooling),Strategic Rationale
Frontend Framework,Next.js 14+,The 2026 industry standard for AI-native applications; provides optimal server-side rendering and edge-runtime compatibility for low-latency LLM interactions.
Language,TypeScript,Enforces strict interface contracts across the RAG pipeline to prevent runtime failures during complex data transformations.
UI Components,Shadcn UI,Delivers an enterprise-standard aesthetic; enables full code ownership and customisation rather than relying on restrictive third-party libraries.
Styling,Tailwind CSS,"Facilitates rapid, maintainable utility-first styling, ensuring UI performance remains high as the platform complexity scales."
Reasoning Model,OpenAI GPT-4.1/5,Selected for superior reasoning capabilities and grounded instruction following; essential for the 11-stage inference path.
Embedding Model,text-embedding-3-small,"Chosen for the optimal balance of cost-efficiency and performance, providing a 1536-dimension vector space for high-precision retrieval."

### UI Performance and Stability: System Font Stack

To achieve enterprise-standard UI performance, we have implemented a 'System Font Stack' resolution. This is a deliberate architectural decision to minimise **Cumulative Layout Shift (CLS)** and achieve a "speed-first" feel. By bypassing custom web font requests, the "Scientific Lab" interface remains stable and responsive, signalling professional technical judgment from the user's first interaction.

## 3. Security Architecture and Database Engineering

The backend infrastructure is powered by Supabase, providing a managed PostgreSQL ecosystem that handles authentication, vector storage, and file management within a single integrated environment.

### Vector Capabilities and Schema Engineering

The pgvector extension has been enabled to support semantic similarity searches. We have implemented the documents table to track file metadata and established the parent-child relationship required for document processing.**Key Schema Fields ( documents table):**

- **id (uuid):** Primary key with uuid_generate_v4().
- **user_id (uuid):** References auth.users(id) for strict data ownership.
- **filename (text):** Original file name for user reference.
- **storage_path (text):** Reference to the physical file in Supabase Object Storage.
- **status (text):** Defaulting to 'uploaded', tracking the ingestion lifecycle.
- **created_at (timestamp):** UTC-timestamped record of ingestion.**Architectural Note:** The schema is designed for data integrity; specifically, the document_chunks table (to be fully populated in Phase 3) utilises on delete cascade logic. This ensures that deleting a parent document automatically purges all associated vector chunks, maintaining a clean and cost-effective database state.

### Security Layer: Multi-Tenant Readiness

Security is enforced via Supabase Auth and the application of Row Level Security (RLS) on the documents table. RLS is not merely used for isolation but is implemented with **multi-tenant architecture readiness** in mind. Every query is filtered at the database level by the user's unique user_id, ensuring that sensitive document data is strictly isolated between users—a non-negotiable requirement for enterprise AI applications.

## 4. UI Architecture: The 'Scientific Lab' Narrative

The interface adopts a "Scientific Lab" aesthetic to move away from the "black box" chatbot anti-pattern. This narrative focuses on technical transparency, exposing the internal mechanics of the RAG pipeline to provide empirical evidence of the system's accuracy.

### Core Layout and Pipeline Visibility

The UI skeleton prioritises visibility into the **11-stage inference and evaluation paths** . Core components include:

- **Sidebar Navigation:** Integrated access to Documents, Settings, and the Evaluation Hub.
- **Landing Page:** A professional entry point for secure authentication.

- **Evaluation Hub (Placeholder):** Designed to eventually visualise real-time Ragas metrics (Faithfulness, Relevance, etc.).
- **Experiment Leaderboard (Placeholder):** A configuration comparison view to prove that settings like chunk size were chosen based on data, not intuition.This architecture ensures that components like retrieved chunks and similarity scores are accessible to the user, providing the "traceability" that senior recruiters prioritise.

## 5. Deployment and CI/CD Strategy

The project follows a "Day 1" deployment philosophy. By hosting on **Vercel**, we ensure CI/CD parity and match the development environment to the live production environment immediately.

### Rationale for Vercel Integration

1. **Production Parity:** Eliminates environment desynchronisation by synchronising environment variables and edge-runtime constraints from the first commit.
2. **Risk Mitigation:** Identifies deployment-specific issues (e.g., build-time errors or secret mismatches) before introducing LLM latency.
3. **CI/CD Pipeline:** Automated deployments from the GitHub repository allow for a continuous, professional engineering workflow.

## 6. Environment Management and Project Specifications

We have standardised the local environment to ensure maintainability and secure collaboration. The .env.local.example file contains placeholder keys for Supabase and OpenAI, allowing future maintainers to synchronise their local environments rapidly and securely.

### Developer Note: Pending Hydration

Maintainers must be cognizant of the 'Pending Hydration' reminder in the project specification. In a Next.js 14+ environment, this is a critical measure to prevent **client-side state desynchronisation**. Given our focus on real-time evaluation metrics, ensuring the UI remains synchronised with server-side logic during initial page load is vital for maintaining UI stability and professional-grade performance.

## 7. Phase 1 Summary and Readiness for Phase 2

Phase 1 has successfully delivered a stable, secure, and deployed infrastructure skeleton.

### Specific Deliverables Achieved

- **Live Production URL:** Hosted on Vercel with integrated CI/CD via GitHub.
- **Secure Auth Flow:** Functional login and signup system using Supabase Auth.
- **Hardened Database:** Secure PostgreSQL instance with pgvector enabled and RLS-protected metadata tables.
- **Responsive Layout:** Enterprise-grade "Scientific Lab" dashboard ready for pipeline visualisation.

## Bridge to Phase 2

The platform is now primed for **Phase 2: Secure File Management**. The established documents table and Supabase storage configuration provide the prerequisite foundation for implementing PDF ingestion, object storage referencing, and the initial stages of the 11-stage RAG pipeline.